

TECHNICAL DESIGN SUBMISSION

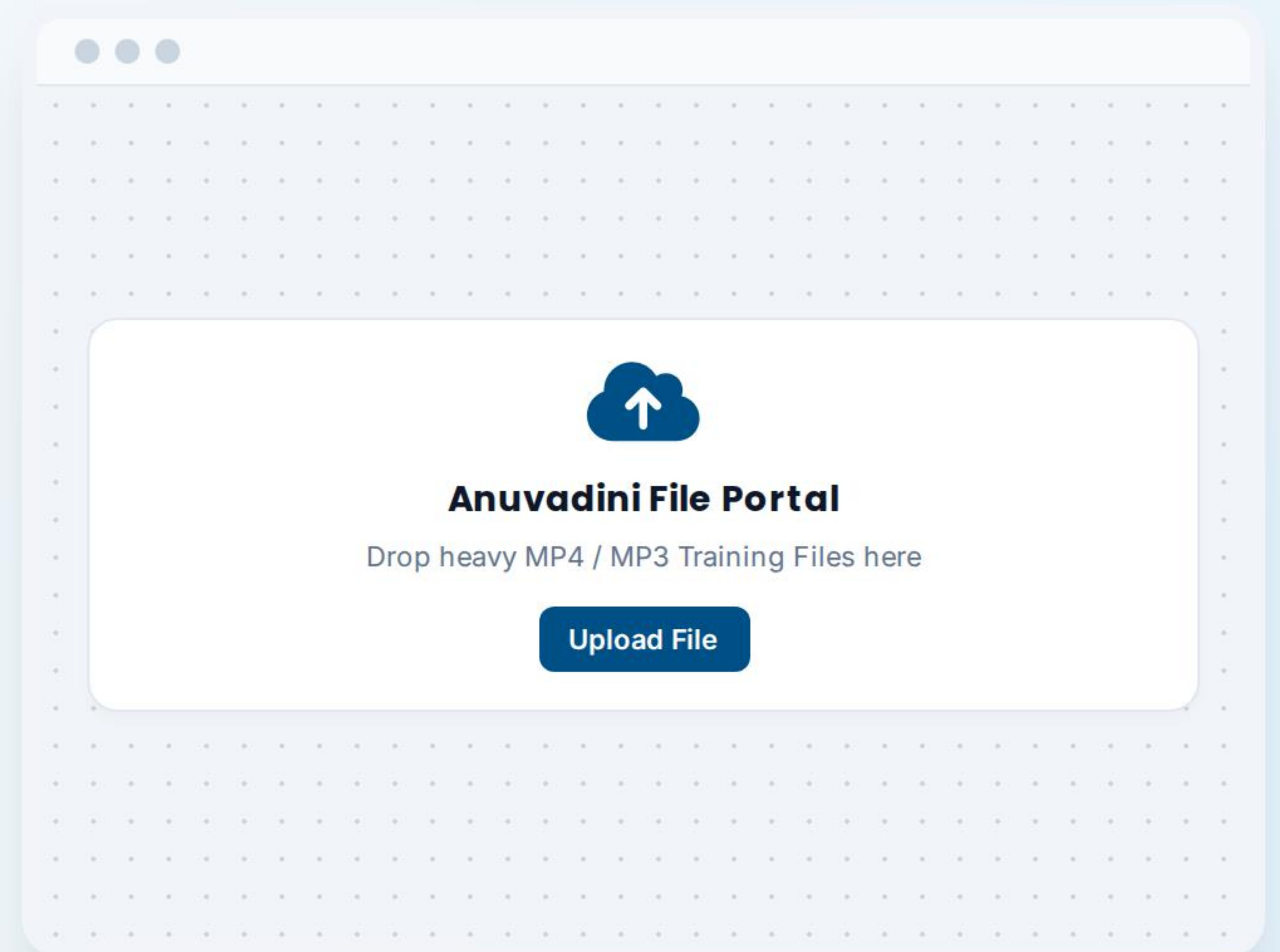
अनुवादिनी (Anuvadini)

An offline-first, CPU-optimized translation suite designed to bring multi-format agricultural training directly to Indian rural communities.

 Powered by HSBC

 Built for BAIF

100% Offline Core



| Problem Statement & Field Context

 Indian farmers working in a green agricultural field

Language Barriers in Rural Extension

Field trainers require rapid localization of audio/video manuals from English & Hindi into regional languages (e.g., **Marathi**) for offline deployment.

🚫 **No-GPU Dependency:** Must run reliably on existing low-spec single-core CPU desktops in Pune/BAIF offices.

Disconnected Field Reality: Internet is only for installation; translations happen strictly offline before traveling to remote regions.

🤖 **Strict Zero-Cost Mandate:** Absolutely zero software/licensing budgets. Must utilize open-source frameworks exclusively.

| अनुवादिनी Core Pipeline Design

The application orchestrates a continuous, localized translation loop running entirely on CPU. It parses inputs, processes them across isolated model layers, and yields synchronized media deliverables.

Step 1

Media Intake

Accepts MP4, MKV, MP3, WAV, etc. segments up to 15 mins. Processes audio extraction using FFmpeg subprocesses.

Step 2

STT Engine

Launches Faster-Whisper (INT8 quantized). Decodes speech with localized dialect filters on single-thread CPU cycles.

Step 3

NMT Translation

AI4Bharat IndicTrans2 engine maps Marathi and Hindi semantics. Quantization offsets memory use below 2GB RAM.

Step 4

Asset Delivery

Compiles localized audio files via VITS TTS, synthesizes SRT/VTT captions, and burns subtitles using local FFmpeg layers.

Proposed Open-Source Tech Stack

Custom quantization transforms demanding LLM/NMT models into lightweight structures executable on base corporate PCs.

Task Category	Open-Source Standard	Offline Optimization Strategy
ASR (Speech-to-Text)	Faster-Whisper (Medium-Quantized)	Intel MKL & CTranslate2 orchestration reduces raw RAM footprint down to ~1.2 GB, boosting execution times.
NMT (Translation)	IndicTrans2 (AI4Bharat)	Optimized 200M/400M parameters distilled weights specifically for regional Marathi syntax and grammatical patterns.
TTS (Audio Synthesis)	VITS / FastPitch Pipeline	Non-autoregressive model that converts text back into natural regional voiceover output with instant CPU rendering.
Media Multiplexing	FFmpeg Pipeline	Compiles and synchronizes translated voice track overlays and burns SRT captions directly onto output MP4s.

| Compliance with Evaluation Pillars (1)



1. Impact & Reach

Ensures vital farming guidance (soil health, animal husbandry) reaches Marathi and Hindi-speaking communities, democratizing agricultural extension.



2. Scalability

Slices dependencies down to single CPU setups. Requires no active GPU clusters, making the software distributable via USB drives to any regional sub-office.



3. Production Readiness

Built-in dynamic audio segmentation allows the app to process files up to 15 minutes without running out of standard RAM or system cache.

| Compliance with Evaluation Pillars (2)



4. Usability / UX

Simplified file portal mimics the provided mockups: clear drag-and-drop mechanics, side-by-side editing review panels, and single-click transcript downloads.



5. Tech Safety

100% on-premises execution guarantees zero data leaks or external network breaches. Preserves native scripts for Marathi and Hindi translation accuracy.



6. Zero Cost (OS)

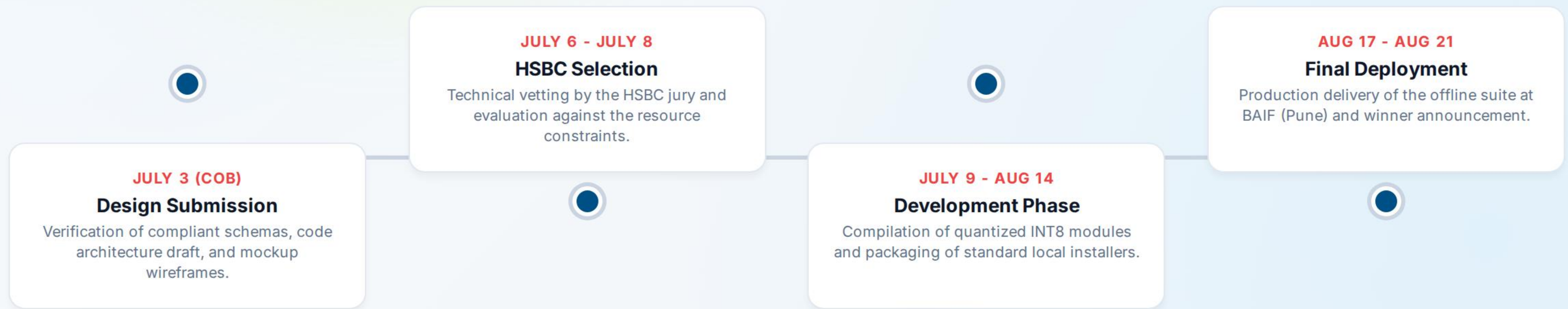
Relies strictly on open-source weights. No paid OpenAI, Microsoft Azure, or AWS cloud APIs. Free of ongoing software licensing overhead.



7. Optimization

Leverages CTranslate2 compilation pipelines and INT8 quantization matrices, ensuring the heaviest translation layers render in real-time.

| Project Implementation Roadmap



| Image Sources



https://images.splitshire.com/full/Ramu-the-Farmer:-A-Kid-Friendly-Animated-Village-Morning_Prre3.png

Source: www.splitshire.com